

Nonlinear Shallow Water Equations and Boundary Conditions for Tide–Storm–Surge Modeling

Abstract

This note describes the nonlinear shallow water equations (SWE) for modeling tide, storm surge in coastal regions. We describe the governing equations, physical source terms, and their interpretation. A positivity-preserving transformation of the water depth is introduced to improve numerical robustness, particularly for implicit solvers and discontinuous Galerkin discretizations. The transformed system is derived and analyzed in a form suitable for high-order numerical methods. The note summarizes boundary conditions commonly used in tide–storm–surge modeling such as open-ocean boundaries, land boundaries, riverine boundaries, internal hydraulic structures, and wetting–drying interfaces. Both physical interpretation and mathematical formulations are presented, with emphasis on practical choices for coastal and inundation simulations.

1 Introduction

Tide–storm–surge modeling is commonly based on the depth-averaged nonlinear shallow water equations (SWE), possibly augmented by atmospheric forcing, bottom friction, Coriolis effects, horizontal viscosity, and wave radiation stresses. The predictive accuracy of such models depends not only on the governing equations and numerical discretization, but also on the proper specification of boundary conditions.

2 Governing Variables

We define the primary variables as follows:

- $\eta(x, y, t)$: free surface elevation,
- $h(x, y)$: bathymetric depth (positive downward),
- $H = h + \eta$: total water depth,
- $\mathbf{u} = (u, v)$: depth-averaged velocity,
- $\mathbf{q} = H\mathbf{u}$: volume flux,
- ρ_w : water density,
- g : gravitational acceleration.

$$\boxed{\begin{aligned} \frac{\partial H}{\partial t} + \nabla \cdot (H\mathbf{u}) &= 0, \\ \frac{\partial(H\mathbf{u})}{\partial t} + \nabla \cdot \left(H\mathbf{u} \otimes \mathbf{u} + \frac{1}{2}gH^2\mathbf{I} \right) &= \mathbf{S}. \end{aligned}} \quad (1)$$

where $\mathbf{S}$ includes bathymetric forcing, atmospheric pressure gradients, wind stress, Coriolis effects, bottom friction, viscous terms, and possibly other source terms as

$$\mathbf{S} = -gH\nabla h - \frac{H}{\rho_w}\nabla p_{atm} + \frac{\boldsymbol{\tau}_w}{\rho_w} + f\hat{\mathbf{k}} \times (H\mathbf{u}) - C_f|\mathbf{u}|(H\mathbf{u}) + \nabla \cdot (\nu\nabla H\mathbf{u}). \quad (2)$$

3 Positivity-Preserving Transformation

To ensure $H \geq 0$ during numerical simulations, we introduce the transformation:

$$H(\phi) = H_{\min} + e^\phi, \quad H_{\min} \geq 0.$$

This guarantees positivity for all $\phi \in \mathbb{R}$.

$$e^\phi \partial_t \phi + \nabla \cdot \mathbf{q} = 0, \quad \Rightarrow \quad \partial_t \phi = -e^{-\phi} \nabla \cdot \mathbf{q}.$$

Using product rule:

$$\partial_t \phi + \nabla \cdot (e^{-\phi} \mathbf{q}) + e^{-\phi} \mathbf{q} \cdot \nabla \phi = 0.$$

$$\boxed{\begin{aligned} \partial_t \phi + \nabla \cdot (e^{-\phi} \mathbf{q}) + e^{-\phi} \mathbf{q} \cdot \nabla \phi &= 0, \\ \partial_t \mathbf{q} + \nabla \cdot \left(\frac{\mathbf{q} \otimes \mathbf{q}}{H_{\min} + e^\phi} + \frac{1}{2}g(H_{\min} + e^\phi)^2\mathbf{I} \right) &= \mathbf{S}. \end{aligned}}$$

where

$$\mathbf{S} = -g(H_{\min} + e^\phi)\nabla h - \frac{(H_{\min} + e^\phi)}{\rho_w}\nabla p_{atm} + \frac{\boldsymbol{\tau}_w}{\rho_w} + f\hat{\mathbf{k}} \times (\mathbf{q}) - C_f\left|\frac{\mathbf{q}}{H_{\min} + e^\phi}\right|(\mathbf{q}) + \nabla \cdot (\nu\nabla \mathbf{q}). \quad (3)$$

The positivity-preserving transformation enhances numerical stability while maintaining consistency with the underlying physics. This formulation is particularly advantageous for high-order discretizations and large-scale GPU-accelerated simulations.

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3.1 Guaranteed Positivity and Physical Admissibility

The transformation

$$H(\phi) = H_{\min} + e^\phi$$

ensures

$$H \geq H_{\min} \geq 0 \quad \text{for all } \phi \in \mathbb{R}.$$

This property eliminates the possibility of negative water depth, which is unphysical and can cause catastrophic failure in standard formulations. In particular:

- No additional limiters or clipping are required to enforce $H \geq 0$.
- The solution remains within the physically admissible set throughout the simulation.
- Numerical breakdown due to negative depth is completely avoided.

3.2 Robust Treatment of Wetting and Drying

Wetting–drying transitions are one of the main challenges in coastal and inundation modeling. The transformation provides a natural and robust framework for handling vanishing water depth:

- As $\phi \rightarrow -\infty$, $H \rightarrow H_{\min}$ smoothly.
- Dry states are approached continuously without introducing discontinuities in the governing variables.
- The shoreline emerges naturally from the solution without requiring explicit front tracking.

This avoids many difficulties associated with traditional wetting–drying algorithms, such as:

- ad hoc thresholding,
- switching between wet and dry states,
- instability near thin water layers.

3.3 Improved Nonlinear Stability

The exponential mapping regularizes the solution space and improves the behavior of nonlinear solvers:

- The transformed system avoids singularities associated with $H \rightarrow 0$ in terms such as $\frac{1}{H}$.
- Nonlinear residuals remain well-defined even in nearly dry regions.
- The mapping introduces a natural barrier that prevents iterates from leaving the admissible region.

This is particularly important for implicit time integration and steady-state computations.

3.4 Advantages for Newton-Type Solvers

For Newton and Newton–Krylov methods, the transformation provides significant benefits:

- The Jacobian is well-defined for all ϕ .
- The derivatives are smooth and simple:

$$\frac{dH}{d\phi} = e^\phi, \quad \frac{d}{d\phi} \left(\frac{1}{H} \right) = -\frac{e^\phi}{H^2}, \quad \frac{d}{d\phi}(H^2) = 2He^\phi.$$

- Line search or trust-region methods are more effective because feasibility ($H \geq 0$) is automatically preserved.

As a result, the nonlinear solver is more robust and less sensitive to initial guesses.

3.5 Consistency with Conservation Laws

The transformation preserves the conservative structure of the shallow water equations:

- The mass conservation equation remains in conservative divergence form.
- Momentum conservation is maintained with modified but consistent fluxes.
- Source terms remain physically interpretable.

Thus, the transformation does not alter the underlying physics, but only reparameterizes the state variables.

3.6 Compatibility with High-Order Methods

The smoothness of the exponential mapping makes it particularly suitable for high-order discretizations such as DG and HDG:

- No need for nonlinear limiters to enforce positivity.
- High-order accuracy is preserved even near wet–dry interfaces.
- The formulation avoids non-smooth switching logic that can degrade convergence.

This is especially important for:

- high-order polynomial approximations,
- hp-adaptive methods,
- spectral accuracy in smooth regions.

3.7 Benefits for GPU and Parallel Computing

The transformation is also advantageous for modern high-performance computing architectures:

- Eliminates conditional branching associated with wet/dry switching.
- Enables uniform kernel execution across elements.
- Improves vectorization and GPU efficiency.
- Reduces divergence in parallel threads.

These properties are critical for scalable implementations on GPUs and distributed systems.

3.8 Smooth Coupling with Boundary Conditions

The positivity-preserving formulation facilitates consistent treatment of boundary conditions:

- Open, wall, and mixed boundary conditions can be expressed smoothly in terms of ϕ .
- Wet/dry dependent boundary treatments can be formulated using $H(\phi)$ without discontinuities.
- Blended boundary conditions can be implemented as smooth functions of ϕ .

3.9 Summary

The positivity-preserving transformation provides a mathematically consistent and numerically robust framework for solving the shallow water equations. Its main advantages include:

- guaranteed positivity of water depth,
- robust handling of wetting–drying,
- improved nonlinear solver performance,
- compatibility with high-order discretizations,
- enhanced efficiency on modern computing architectures.

These properties make the transformation particularly well suited for large-scale, high-fidelity tide–storm–surge simulations involving complex coastal geometries and inundation processes.

4 Boundary Conditions

Boundary conditions in tide–storm–surge modeling are diverse because coastal systems involve multiple physical processes and multiple types of interfaces. Open-ocean boundaries must allow tidal forcing and outgoing wave radiation. Land boundaries may represent impermeable coastlines, rough walls, or engineered structures such as levees and gates. Riverine boundaries may require discharge or stage conditions. Wetting–drying interfaces are generally treated as moving internal fronts rather than fixed external boundaries. In complex domains, mixed and state-dependent boundary conditions are often necessary. A careful and physically consistent choice of boundary conditions is therefore essential for accurate coastal, estuarine, and inundation modeling. In tide–storm–surge modeling, boundary conditions are usually grouped into the following categories:

1. Open-ocean boundaries
2. Land boundaries
3. River and discharge boundaries
4. Hydraulic structure boundaries
5. Wetting–drying interfaces
6. Mixed and state-dependent boundaries
7. Pump and drainage boundaries
8. Atmospheric and Wave-Coupled Boundary Effects

5 Open-Ocean Boundary Conditions

Open-ocean boundaries represent artificial truncations of the sea domain through which tidal waves, storm surge signals, and long gravity waves may enter or leave the computational domain.

5.1 Prescribed Free-Surface Elevation

The simplest open boundary condition prescribes the free-surface elevation:

$$\eta = \eta_{\text{ext}}(x, y, t) \quad \text{on } \Gamma_{\text{open}}. \quad (4)$$

Equivalently,

$$H = h + \eta_{\text{ext}}. \quad (5)$$

This condition is widely used when tidal harmonics or externally generated surge signals are available. A common representation is

$$\eta_{\text{ext}}(t) = \sum_{k=1}^{N_c} A_k \cos(\omega_k t + \phi_k), \quad (6)$$

where A_k , ω_k , and ϕ_k are the amplitudes, frequencies, and phases of tidal constituents.

This approach is simple and effective, but it may generate spurious reflections if used without proper radiation treatment.

5.2 Prescribed Normal Discharge or Velocity

Another possibility is to prescribe the normal discharge or normal velocity:

$$(H\mathbf{u}) \cdot \mathbf{n} = q_n^{\text{ext}}(x, y, t) \quad \text{on } \Gamma_{\text{open}}, \quad (7)$$

or

$$\mathbf{u} \cdot \mathbf{n} = u_n^{\text{ext}}(x, y, t). \quad (8)$$

This type of condition is appropriate when inflow or outflow through the boundary is known from external data or coupled models.

5.3 Characteristic Boundary Conditions

Because the shallow water equations are hyperbolic, physically consistent open boundaries should distinguish between incoming and outgoing waves. In the normal direction, the characteristic variables are approximately

$$R^\pm = u_n \pm \frac{\eta}{\sqrt{gH}}, \quad u_n = \mathbf{u} \cdot \mathbf{n}. \quad (9)$$

At an open boundary, one typically prescribes only the incoming characteristic and allows the outgoing characteristic to be determined by the interior solution. Symbolically,

$$R^- = R_{\text{ext}}^-, \quad R^+ = R_{\text{int}}^+. \quad (10)$$

This is the most physically appropriate framework for radiation-type open boundaries.

5.4 Flather Boundary Condition

A widely used radiation condition is the Flather condition:

$$u_n = u_{n,\text{ext}} + \frac{\eta - \eta_{\text{ext}}}{\sqrt{gH}}. \quad (11)$$

Here η_{ext} and $u_{n,\text{ext}}$ come from external tidal forcing, large-scale ocean models, or climatological data. The Flather condition allows outgoing waves to leave the domain while still imposing large-scale incoming tidal information.

5.5 Sommerfeld or Radiation Boundary Condition

A simpler radiative condition may be written in the form

$$\frac{\partial \eta}{\partial t} + c \frac{\partial \eta}{\partial n} = 0, \quad c = \sqrt{gH}, \quad (12)$$

which approximates one-way wave propagation normal to the boundary. Such conditions are useful for reducing artificial reflections, but they are generally less accurate than characteristic or Flather-type formulations when strong tidal forcing must also be imposed.

5.6 Combined Tide and Surge Boundary Condition

In many practical models, the open boundary signal is decomposed into astronomical tide and meteorological surge:

$$\eta_{\text{ext}} = \eta_{\text{tide}} + \eta_{\text{surge}}. \quad (13)$$

The boundary condition may then prescribe total elevation, or use η_{ext} and $u_{n,\text{ext}}$ in a characteristic formulation.

6 Land Boundary Conditions

Land boundaries represent coastlines, shorelines, seawalls, cliffs, harbor walls, levees, and other impermeable or partially permeable structures.

6.1 Impermeable Slip Wall

The most common land boundary condition is the no-normal-flow condition

$$\mathbf{u} \cdot \mathbf{n} = 0 \quad \text{on } \Gamma_{\text{land}}. \quad (14)$$

Equivalently,

$$(H\mathbf{u}) \cdot \mathbf{n} = 0. \quad (15)$$

This condition models an impermeable boundary while allowing tangential motion. It is the standard choice for rigid coastlines in depth-averaged models.

6.2 No-Slip Wall

A more restrictive condition is

$$\mathbf{u} = \mathbf{0} \quad \text{on } \Gamma_{\text{land}}. \quad (16)$$

This imposes both zero normal velocity and zero tangential velocity. It is rarely used in depth-averaged storm-surge models because bottom friction usually dominates over lateral wall shear, but it may be useful in special applications or highly idealized studies.

6.3 Partial-Slip or Frictional Land Boundary

A more general land boundary condition can combine impermeability with tangential drag:

$$\mathbf{u} \cdot \mathbf{n} = 0, \quad (17)$$

together with a tangential stress law such as

$$\tau_t = -C_t |\mathbf{u}_t| \mathbf{u}_t, \quad (18)$$

where $\mathbf{u}_t$ is the tangential component of velocity and C_t is an empirical drag coefficient.

This type of condition can be used to represent enhanced resistance along urban or rough coastal boundaries.

7 River and Discharge Boundary Conditions

Riverine boundaries are important when coastal surge interacts with river discharge, estuarine inflow, or controlled outflow.

7.1 Prescribed Inflow Discharge

One may prescribe a total river discharge $Q(t)$ distributed along a river boundary Γ_{river} :

$$\int_{\Gamma_{\text{river}}} (H\mathbf{u}) \cdot \mathbf{n} ds = -Q(t), \quad (19)$$

where the negative sign indicates inflow into the domain if $\mathbf{n}$ is the outward normal.

In simpler settings, a pointwise discharge condition may be imposed:

$$(H\mathbf{u}) \cdot \mathbf{n} = q_n^{\text{river}}(x, y, t). \quad (20)$$

7.2 Prescribed Stage Boundary

In some river models, the water surface elevation is prescribed instead:

$$\eta = \eta_{\text{river}}(t) \quad \text{on } \Gamma_{\text{river}}. \quad (21)$$

7.3 Rating-Curve Boundary Condition

A stage–discharge relation may also be used:

$$Q = Q(\eta), \quad (22)$$

or equivalently

$$(H\mathbf{u}) \cdot \mathbf{n} = f(\eta), \quad (23)$$

which is useful for hydraulic control at river mouths or engineered systems.

8 Hydraulic Structure Boundary Conditions

Storm-surge models often include levees, seawalls, culverts, gates, barriers, and weirs. These are not standard walls or open boundaries, but internal or boundary interfaces governed by hydraulic laws.

8.1 Weir or Overtopping Boundary Condition

A classical overtopping law is

$$q_n = C_w \max(0, \eta_1 - \eta_2 - z_c)^{3/2}, \quad (24)$$

where z_c is the crest elevation, C_w is a discharge coefficient, and $\eta_1 - \eta_2$ is the water-level difference across the structure.

This condition allows flow only when the upstream water level exceeds the crest elevation.

8.2 Orifice or Culvert Boundary Condition

Flow through a submerged opening may be modeled by an orifice law such as

$$q_n = C_o A \sqrt{2g |\Delta\eta|} \text{sign}(\Delta\eta), \quad (25)$$

where A is the opening area and $\Delta\eta$ is the head difference.

8.3 Gate-Control Boundary Condition

For barriers or gates, the discharge may depend on gate opening and head difference:

$$q_n = q_n(\Delta\eta, \text{gate opening}, t). \quad (26)$$

These conditions are usually implemented as nonlinear flux laws rather than classical Dirichlet or Neumann conditions.

9 Wetting–Drying and Moving Shoreline

The wetting–drying interface is not a conventional external boundary condition. Instead, it is a moving internal transition between wet cells and dry cells.

9.1 Wetting–Drying Concept

Let

$$H = h + \eta.$$

A point is considered wet if

$$H > H_{\text{tol}}, \quad (27)$$

and dry if

$$H \leq H_{\text{tol}}, \quad (28)$$

where H_{tol} is a small threshold.

9.2 Numerical Treatment

In inundation modeling, the shoreline is not imposed as a fixed boundary. Instead, one includes potentially dry land within the computational domain and uses a numerical wetting–drying treatment that ensures

$$H \geq 0. \quad (29)$$

Common strategies include:

- positivity-preserving reconstructions,
- flux limiting near dry cells,
- special Riemann solvers for wet–dry fronts,
- transformed variables such as

$$H(\phi) = H_{\min} + e^{\phi}.$$

10 Mixed and State-Dependent Boundary Conditions

In practical coastal domains, some artificial boundaries may intersect regions that are wet at some times and dry at others. In such cases, a boundary segment cannot be permanently classified as purely open or purely land.

10.1 State-Dependent Partition

Let Γ_b be a boundary segment and define

$$\Gamma_b^{\text{wet}}(t) = \{\mathbf{x} \in \Gamma_b : H(\mathbf{x}, t) > H_{\text{tol}}\}, \quad (30)$$

$$\Gamma_b^{\text{dry}}(t) = \{\mathbf{x} \in \Gamma_b : H(\mathbf{x}, t) \leq H_{\text{tol}}\}. \quad (31)$$

Then one may impose

$$\mathcal{B} = \mathcal{B}_{\text{open}} \quad \text{on } \Gamma_b^{\text{wet}}(t), \quad \mathcal{B} = \mathcal{B}_{\text{wall}} \quad \text{on } \Gamma_b^{\text{dry}}(t). \quad (32)$$

10.2 Blended Boundary Condition

To avoid abrupt switching, one may define a wetness function $\alpha(H) \in [0, 1]$, for example

$$\alpha(H) = \begin{cases} 0, & H \leq H_{\text{dry}}, \\ \frac{H - H_{\text{dry}}}{H_{\text{wet}} - H_{\text{dry}}}, & H_{\text{dry}} < H < H_{\text{wet}}, \\ 1, & H \geq H_{\text{wet}}, \end{cases} \quad (33)$$

and blend open and wall boundary fluxes:

$$\widehat{F}_n = \alpha(H) \widehat{F}_n^{\text{open}} + (1 - \alpha(H)) \widehat{F}_n^{\text{wall}}. \quad (34)$$

This strategy is particularly attractive for implicit methods and HDG or DG discretizations because it provides a smoother transition between wet and dry states.

11 Pump and Drainage Boundary Conditions

Pumping stations and drainage systems represent actively controlled fluxes between the computational domain and external storage or receiving waters.

A pump boundary condition may be written as

$$(H\mathbf{u}) \cdot \mathbf{n} = -Q_{\text{pump}}(t, \eta), \quad (35)$$

where Q_{pump} is a prescribed or state-dependent discharge.

In many cases, the pump discharge is limited by capacity:

$$Q_{\text{pump}} = \min(Q_{\text{max}}, f(\eta_{\text{in}}, \eta_{\text{out}})). \quad (36)$$

A simple on/off control law is

$$Q_{\text{pump}} = \begin{cases} Q_{\text{max}}, & \eta_{\text{in}} > \eta_{\text{on}}, \\ 0, & \eta_{\text{in}} < \eta_{\text{off}}. \end{cases} \quad (37)$$

Drainage systems may also be modeled as distributed sinks:

$$\frac{\partial H}{\partial t} + \nabla \cdot (H\mathbf{u}) = -S_{\text{drain}}, \quad (38)$$

where S_{drain} represents removal of water due to infiltration or engineered drainage networks.

12 Atmospheric and Wave-Coupled Boundary Effects

Although atmospheric pressure and wind stress are usually applied as body-force source terms rather than boundary conditions, some coupled tide–storm–surge models also incorporate external wave forcing or wave radiation stress effects through boundary information.

Examples include:

- prescribed incoming wave setup,
- wave-averaged radiation stress contributions near open boundaries,
- coupling to larger-scale ocean, wave, or atmospheric models.

These are usually represented through externally prescribed η_{ext} , $u_{n,\text{ext}}$, or additional boundary flux terms.

13 Summary of Main Boundary Conditions

Table 1 summarizes the most common boundary conditions used in tide–storm–surge models.

For many tide–storm–surge applications, the following choices are appropriate:

- **Open-ocean boundaries:** characteristic or Flather-type conditions with prescribed tidal elevation.
- **Rigid coastlines:** impermeable slip-wall condition.
- **River boundaries:** prescribed discharge, prescribed stage, or rating curve.
- **Levees and barriers:** nonlinear hydraulic flux laws.
- **Flooding and inundation:** wetting–drying treatment within the computational domain.
- **Artificial mixed wet/dry boundaries:** blended state-dependent boundary fluxes.

Table 1: Summary of common boundary conditions in tide–storm–surge modeling.

Boundary type	Mathematical form	Typical use
Open elevation	$\eta = \eta_{\text{ext}}$	Tidal forcing
Open discharge	$(H\mathbf{u}) \cdot \mathbf{n} = q_n^{\text{ext}}$	Known inflow/outflow
Characteristic	$R^- = R_{\text{ext}}^-$	Radiation + incoming tide
Flather	$u_n = u_{n,\text{ext}} + \frac{\eta - \eta_{\text{ext}}}{\sqrt{gH}}$	Coastal/open sea
Slip wall	$\mathbf{u} \cdot \mathbf{n} = 0$	Impermeable coast
No-slip wall	$\mathbf{u} = \mathbf{0}$	Rare special cases
Frictional wall	$\mathbf{u} \cdot \mathbf{n} = 0, \tau_t = -C_t \mathbf{u}_t \mathbf{u}_t$	Rough boundary
River inflow	$\int (H\mathbf{u}) \cdot \mathbf{n} ds = -Q(t)$	River discharge
Weir/levee	$q_n = C_w \max(0, \eta_1 - \eta_2 - z_c)^{3/2}$	Flood barriers
Wet–dry interface	$H \geq 0$ with wetting–drying treatment	Inundation
Mixed wet/dry	$\hat{F}_n = \alpha \hat{F}_n^{\text{open}} + (1 - \alpha) \hat{F}_n^{\text{wall}}$	State-dependent boundary